

Choi Taewoong

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Education

Sungkyunkwan University 2021 – Present
B.S. in Mechanical Engineering GPA : 3.67 / 4.50

Research Experience

POSTECH HERO Lab Jul 2026 – Present
Undergraduate Research Intern
Conducting research on underwater robotic systems, with a focus on bio-inspired mechanism design, robot dynamics, and hydrodynamic analysis.

SKKU Robotics Innovatory Jan 2025 – May 2026
Undergraduate Research Intern
Conducted research on autonomous mobile robots, focusing on LiDAR–Radar sensor fusion and mobile robot control.

Awards

Grand Prize, SKKU Creative Capstone Design Competition
AI-Based Directional Audio Guidance System for Pedestrians
Developed an assistive intersection guidance system that detects a pedestrian's help-request gesture using YOLOv8-Pose, tracks the selected user in real time with a two-axis actuation mechanism, and delivers directional audio only toward the target. Received the Grand Prize and qualified to advance to the national-level competition.



Bronze Prize, SKKU Creative Capstone Design Competition
Stair-Climbing Disaster-Reconnaissance Rover
Received the Bronze Prize for developing a rocker–bogie rover with transformable wheels for stair climbing; served as team leader of a ten-member team.



Scholarship

Future Mobility Bootcamp Scholarship 2026
Awarded for studies in future mobility and automotive electrification , KRW 3,600,000

Other Experience

SKKU Makerspace Manager Dec 2025 – Present
3D Printing and Laser Cutting Manager
Operate and maintain 3D printers and laser-cutting equipment and support users with equipment setup, and safe machine operation.

Republic of Korea Air Force Feb 2023 – Nov 2024
Mandatory Military Service, Honorably Discharged

Projects

LiDAR–Radar Sensor Fusion for Reliable Navigation Aug 2025 – Mar 2026

Co-op Project, SKKU Robotics Innovatory

Developed a Graph Attention Network to estimate uncertainty in LiDAR and radar measurement and integrated the estimated uncertainty into a Dynamic Occupancy Grid Map for reliable obstacle detection.



Rocker–Bogie Rover with Transformable Wheels Mar 2026 – Jun 2026

SKKU "Creative Capstone Design Project" Class, Team Leader

Led a six-member team in developing a disaster-reconnaissance rover featuring a rocker–bogie suspension and four-bar-linkage transformable wheels for enhanced stair-climbing capability.



Kinematics-Based real time Control of a 4WIS Mobile Robot Jul 2025 – Aug 2025

Undergraduate Research Program, SKKU Robotics Innovatory

Developed several driving modes for a four-wheel-independent-steering mobile robot. Implemented Ackermann-based steering kinematics to calculate per-wheel steering angles and velocities, and validated the control architecture through Gazebo simulations and real-robot experiments.



Bio-Inspired Underwater Leg Mechanism Jul 2026 – Present

Undergraduate Research Internship, POSTECH HERO Lab

Designing a 3DOF bio-inspired underwater robotic leg and analyzing its kinematics, dynamics, hydrodynamic forces, and actuator torque requirements.

Bartender Robot Arm Aug 2025 – Nov 2025

Team Project, Team Leader, SKKU Robotics Club

Led the design and fabrication of a four-DOF robotic arm, implemented motion planning and manipulation using ROS 2 and MoveIt, and developed a web-based beverage ordering interface.



Autonomous Basketball-Shooting Robot May 2025 – Jun 2025

Term Project, SKKU “Design Lab on Vibration and Dynamic Systems” class

Developed an autonomous basketball-shooting robot. Designed a launch mechanism that adjusted the shooting angle according to the target distance and implemented autonomous navigation for target approach and shooting.



Real-Time Underwater Crack Detection System Jul 2026 – Present

PKRC Project, POSTECH HERO Lab

Developing a YOLO-based vision pipeline for real-time underwater crack detection on an NVIDIA Jetson platform, including live visualization and automatic storage of detected crack frames.



Gear Train Design for a Quadruped Robot Leg Sep 2022 – Nov 2022

Team Project, SKKU Robotics Club

Designed and modeled the gears and transmission components of a quadruped robot leg, considering the required reduction ratio, joint layout, and mechanical integration.



Autonomous Arduino RC Car Jun 2022

2022 Smart Car Autonomous Driving Competition

Designed and built an Arduino-based autonomous RC car with ultrasonic sensor and infrared sensor.

